

NFR-SEC-01: Security (GDPR & Access)

- **Description:** The system shall protect sensitive medical data and ensure authorized access only.
- **Measurable Metric:** 100% of backend API endpoints handling clinical data must require an active security token.

E.g. When a doctor logs into the frontend with their password, the backend hands them a temporary digital ID badge called a JWT (JSON Web Token)

0% of unmasked patient names shall be sent to external AI APIs.

E.g. [PATIENT_A] has a severe migraine today. Prescribed Ibuprofen.

NFR-PERF-01: Performance & Response Time

- **Description:** The system shall provide rapid AI generation and UI responsiveness.
- **Measurable Metric:** The AI must return a fully structured medical report in **< 5 seconds**.

E.g. use "Streaming" (where the text appears on the screen word-by-word as the AI generates it, just like ChatGPT does) so the doctor doesn't feel like they are waiting.

The Patient Check-in UI must load in **< 2 seconds**.

E.g. Ensure the check-in page doesn't load a bunch of heavy, unnecessary data in the background.

NFR-USAB-01: Usability

- **Description:** The system must be intuitive for non-technical clinical staff.
- **Measurable Metric:**
 1. **For the Receptionist:** Must be able to reschedule an existing appointment in **< 3 clicks**.
 2. **For the Doctor:** Must be able to initiate the AI report generation in exactly **1 click** after entering their shorthand notes.
 3. **For the Patient:** Must be able to complete the mobile QR self-check-in process in **< 4 screens/steps** without asking the receptionist for help.

NFR-REL-01: Reliability & Availability

- **Description:** The system must remain highly available during clinic operating hours and protect against data loss.
- **Measurable Metric:** The cloud servers must maintain 99.9% uptime during business hours (07:00–19:00). In the event of a crash, automated database backups must be fully restorable within 4 hours with zero data lost from the previous day.

The app will be offline for roughly 43 minutes every month.

E.g. Using a cloud provider like AWS or Azure to automatically back up the database every night at 2:00 AM.(technically had 43 minutes of "downtime," but the user never even noticed!)